

THE KARUSH-KUHN TUCKER NECESSARY CONDITIONS

Lecture 27: Presented by Jacomine Grobler



Stellenbosch Unit for Operations Research in Engineering
Department of Industrial Engineering

Constrained analytic optimisation

In previous chapters, we considered unconstrained optimisation problems over infinite space. In this chapter, we consider optimisation problems of a bounded or constrained nature.

We consider a non-linear optimisation problem in which the objective is to

$$\text{maximise } z = f(\mathbf{x})$$

subject to a set of inequality constraints of the form

$$g_j(\mathbf{x}) \leq a_j, \quad j = 1, \dots, p,$$

together with the domain constraint

$$\mathbf{x} \in \mathbb{R}^n,$$

where p is any natural number, the functions f and $g_1, \dots, g_p$ are all twice differentiable with respect to all pairs of their argument components, and $a_1, \dots, a_p$ are real constants.

Any equality constraint of the form

$$g(\mathbf{x}) = a$$

can be rewritten as two simultaneous inequality constraints of the form

$$\begin{aligned} g(\mathbf{x}) &\leq a, \\ -g(\mathbf{x}) &\leq -a. \end{aligned}$$

Similarly, if some variable x_i is required to be non-negative, then the inequality $-x_i \leq 0$ can be included. Moreover, the fact that we consider only the option of maximisation is without loss of generality, since the minimisation of any function $f(\mathbf{x})$ yields the same local optimum locations as does maximising the function $-f(\mathbf{x})$.

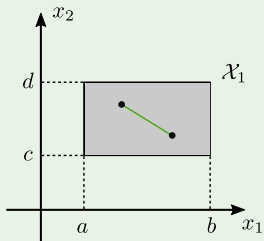
We open this section by extending the notion of convexity from functions to sets. The notion of convexity of a set plays an important role in the theory of non-linear optimisation.

A set $\mathcal{X} \subseteq \mathbb{R}^n$ is *convex* if, for any two elements $\mathbf{x}$ and $\mathbf{x}'$ of $\mathcal{X}$, the straight line segment joining $\mathbf{x}$ and $\mathbf{x}'$ lies entirely within $\mathcal{X}$.

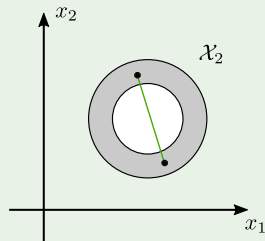
Expressed mathematically, $\mathcal{X}$ is therefore convex if, for any $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ and all $\alpha \in [0, 1]$, it holds that $\mathbf{x}'' = \alpha\mathbf{x} + (1 - \alpha)\mathbf{x}' \in \mathcal{X}$.

If $\alpha = 0$, the point $\mathbf{x}''$ coincides with $\mathbf{x}'$, while if $\alpha = 1$, the point $\mathbf{x}''$ coincides with $\mathbf{x}$. As the parameter α increases from 0 to 1, the point $\mathbf{x}''$ therefore moves along a straight line from $\mathbf{x}'$ to $\mathbf{x}$.

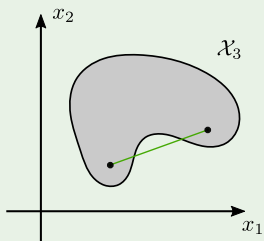
Example 5.1 (p. 165)



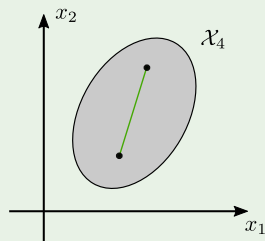
(a) Convex



(b) Not convex



(c) Not convex



(d) Convex

The Karush-Kuhn Tucker necessary conditions

Theorem 5.1 (Karush-Kuhn-Tucker necessary conditions for local optimality) If $\mathbf{x}^*$ is a local optimum of (5.2)–(5.4), then $\mathbf{x}^*$ satisfies the conditions

$$\left. \frac{\partial f}{\partial x_i} \right|_{\mathbf{x}=\mathbf{x}^*} - \sum_{j=1}^p \lambda_j \left. \frac{\partial g_j}{\partial x_i} \right|_{\mathbf{x}=\mathbf{x}^*} = 0, \quad i = 1, \dots, n, \quad (5.5)$$

$$\lambda_j [a_j - g_j(\mathbf{x}^*)] = 0, \quad j = 1, \dots, p, \quad (5.6)$$

$$\lambda_j \geq 0, \quad j = 1, \dots, p. \quad (5.7)$$

The coefficient λ_j in Theorem 5.1 may be interpreted as a *shadow price* associated with the j -th constraint in (5.3), in the sense that if the right-hand side of the j -th constraint in (5.3) were to be increased from a_j to $a_j + \Delta$, for some small value Δ , then the optimal objective function value in (5.2) would be expected to increase by approximately $\Delta\lambda_j$.

The Karush-Kuhn Tucker necessary conditions

The conditions in (5.6) may be considered a generalisation of the well-known complementary slackness conditions in the theory of linear programming, in the following sense:

- If $\lambda_j > 0$, then the j -th constraint is binding (that is, $g_j(\mathbf{x}^*) = a_j$) at a local optimum $\mathbf{x}^*$, while
- if $g_j(\mathbf{x}^*) < a_j$ at a local optimum $\mathbf{x}^*$, then $\lambda_j = 0$.

Theorem 5.2 (Second-order sufficient conditions for global optimality) If the function f in (5.2) is concave and if all the functions $g_1, \dots, g_p$ in (5.3) are convex, then any point $\mathbf{x}^*$ satisfying the Karush-Kuhn-Tucker necessary conditions for local optimality in Theorem 5.1 is a global maximum of (5.2)–(5.4).

Example 5.2 (pp. 168–169)

Determine the dimensions of a customised fish tank with a square bottom and an open top of largest possible enclosed volume, subject to the constraint of using at most four square metres of glass. Suppose the floor length of the tank is x_1 and that its height is x_2 .

